

Question Paper

Exam Date & Time: 13-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH. (COMPUTER SCIENCE & ENGINEERING) DEGREE EXAMINATIONS - JUNE 2025
SUBJECT: CSE 2223 / CSE_2253 EMBEDDED SYSTEMS

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Distinguish between packed and unpacked BCD numbers with suitable examples. Write the steps and code snippet for converting packed BCD number 0x17 to its equivalent ASCII format. (5)
- 1B) Explain the short jump operation in ARM assembly language programming. Discuss their limitations and the method to overcome those limitations. (3)
- 1C) With a suitable example, distinguish between TST and TEQ instructions. (2)
- 2A) Compare the four different types of stack structures with a neat diagram and appropriate example. (4)
- 2B) Identify and explain the assembler directives used in each of the following instructions: (3)
- i) VAL1 RN R1
 - ii) LONG_VAR SPACE 4
 - iii) NUM1 DCD 0x5369
- 2C) Develop an embedded C program for the LPC1768 microcontroller to read the state of a switch (SW) connected to pin P0.1 and display 8-bit even or odd numbers on LEDs connected to pins P1.1 through P1.8. When the switch is pressed, the LEDs should display even numbers; otherwise, they should display odd numbers. (3)
- 3A) i) How many single-edge and double-edge PWM outputs can be generated using the LPC1768 PWM module? (5)
- ii) Explain the concept of shadow registers in the LPC1768 PWM module.
- iii) Draw the timing diagram assuming the PWM period is 6 microseconds and the Timer Counter (TC) increments every 1 microsecond. Let PWM1MR1 = 3 and PWM1MR2 = 4. Whenever the TC value matches any of the match registers, the corresponding PWM output is driven low and remains low until the beginning of the next PWM cycle.
- 3B) Develop an embedded C program using LPC1768 to generate ramp waveform. The Aout pin is P0.26 which should be configured with function 2 (second alternate function). (3)
- 3C) Write Embedded C statements to perform the following: (2)
- i) Configure **P1.16 to P1.20** as input and **P1.24 to P1.29** as output.
 - ii) Set **P0.3** while keeping other bits unchanged.
- 4A) Compose a LPC1768 compliant program to configure Timer 0 with a pre-scaler value of 23 and a match value of 1000, which gives a timer frequency of 1 kHz to generate an interrupt every time the timer reaches the match value set in MR0 for an application that operates in a low power mode. Discuss the steps for the required register setting. (4)
- 4B) Discuss match registers and capture registers of the LPC1768 timer. If the PCLK frequency is 8 MHz, what would be the prescaler value required to achieve a resolution of 3 microseconds? (3)
- 4C) Develop an embedded C program to demonstrate the function of the ADC in the LPC1768 microcontroller. Display the digital output of ADC in LCD assuming the functions related to LCD are available in LCDfunc.h. (3)

- 5A) Draw a flowchart to calculate and set the baud rate divisor for UART communication in the LPC1768 microcontroller. Your flowchart should include the configuration of the Fractional Divider Register (FDR) to improve baud rate accuracy. Assume the main oscillator frequency is 14.7456 MHz and the desired baud rate is 9600. (5)
- 5B) With respect to external interrupts in LPC1768, highlight the importance of EXTINT, EXTMODE and EXTPOLAR registers. Mention the steps involved to configure the Interrupts. (3)
- 5C) A sensor connected to the EINT2 pin of an LPC1768 produces a pulse. The developer wants the microcontroller to trigger an interrupt at the *end* of this pulse (assuming the pulse goes high and then low). Describe the necessary configuration for the EXTMODE and EXTPOLAR registers to achieve this specific triggering behavior. (2)

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Event Name	Subject
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MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025

CSE-2253:Embedded Systems

#	Course Outcome	Description	Marks
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No records found / CO Level Description not defined

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
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